KAZI EHSANUL KARIM

Molecular Simulation | Statistical Mechanics | Multiscale Fluid Mechanics | Machine-Learning Interatomic Potentials

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Postdoctoral / Research Fellow Opportunities | Target start: early 2027

RESEARCH PROFILE

Computational researcher and Ph.D. candidate in Mechanical Engineering at the University of Ulsan, with three first-author articles published in Physical Review E and Physical Chemistry Chemical Physics. Combines molecular dynamics, statistical mechanics and continuum modelling to establish where continuum descriptions of nanoscale transport hold, where they fail and which molecular details they must retain. Delivers results in the form a continuum solver uses: a wall law with a test of where it stops being reliable. Independent work develops error-aware machine-learned interatomic potentials for CNT-confined NaCl–water. Applications include membrane separation, nanofluidic devices, energy conversion and interfacial thermal management; the longer-term aim is adaptive simulation guided by model errors and uncertainty.

Research methods: LAMMPS EMD/NEMD; CP2K; DP-GEN / DeePMD-kit; Green–Kubo transport; continuum boundary conditions; Python / Linux / HPC; ANSYS thermo-fluids.

EDUCATION

University of Ulsan, Republic of Korea Mar 2019 – Present

Ph.D. Candidate, Integrated M.S.–Ph.D. in Mechanical Engineering | Micro/Nano-scale Thermo-Fluids Engineering Laboratory | Supervisor: Prof. BoHung Kim

Ph.D. defense scheduled: 10 December 2026 | Formal degree conferral expected: February 2027

Dissertation: What Survives Sub-nm Confinement? — The Meaning of Slip Length under Molecular Accessibility, Ergodicity, and Structure–Transport Decoupling.

Built on seven studies: three published, two in submission and two in preparation (items 1–7 under Publications & Manuscripts).

Khulna University of Engineering & Technology (KUET), Bangladesh May 2015

B.Sc., Mechanical Engineering

DOCTORAL RESEARCH EXPERIENCE

Doctoral Researcher — Molecular Simulation & Nanofluidics, University of Ulsan Mar 2019 – Present

- **Boundary thermodynamics —** Developed a first-law energy-balance analysis of the nanoscale flow boundary and tested NEMD against Hagen–Poiseuille limits under controlled wall–fluid interaction (Physical Review E, 2021).
- **Water transport in CNTs —** Critically assessed scale effects and boundary definitions in CNT water-transport models, relating molecular-simulation evidence to continuum predictions (Physical Chemistry Chemical Physics, 2024).
- **Confinement and transport —** Analysed molecular admission, packing and transport in subnanometre graphene slits (Physical Chemistry Chemical Physics, 2026); investigate finite-time diffusivity, hydraulic resistance and continuum-closure limits in carbon nanotubes and graphene nanoslits.
- **Reproducibility and uncertainty —** Apply ensemble, timestep, box-size, forcing and sampling checks; block averaging and confidence intervals; and thermostat-sensitivity and cross-method comparisons to test transport interpretations.

INDEPENDENT RESEARCH PROJECT

Error-Aware MLIPs for CNT-Confined NaCl–Water 2026 – Present

- **Workflow —** Built and ran an end-to-end workflow for NaCl–water in four carbon-nanotube diameters — CP2K (PBE-D3(BJ)) ab initio MD seeds and reference labels → DP-GEN → DeePMD-kit → LAMMPS — with a verification step at every stage.
- **Evaluation —** Developed species-resolved force validation on fresh-DFT holdout frames; identified a species-specific failure mode that the aggregate error concealed and committee disagreement did not flag; tested targeted reference-data augmentation (error unchanged; cause under investigation).
- **Transport —** Computed equilibrium (Green–Kubo) interfacial friction from MLIP-driven MD with block-bootstrap uncertainty and model-disagreement screening of trajectories; two standard estimators disagree, and the cause is under investigation.
- **Tooling —** Wrote a tested Python tool-kit for per-species force metrics, group-aware uncertainty, model-scope checks and rules-based decisions; public methods demonstration (synthetic data) at kazi21.github.io.
- **Scientific software —** Developed browser-based molecular and continuum transport laboratories: a 3D inspector of the four ab initio MD trajectories, LAMMPS/WebAssembly water dynamics and a finite-volume ion-transport solver, with numerical verification, traceable molecular inputs and reproducible experiment exports.

PUBLICATIONS & MANUSCRIPTS

Peer-reviewed journal articles

1. K. E. Karim, B. Kim, "First law of thermodynamics on the boundary for flow through a carbon nanotube," Physical Review E, 2021, 103(5), 053115. DOI

2. K. E. Karim, M. Barisik, C. Bakli, B. Kim, "Estimating water transport in carbon nanotubes: A critical review and inclusion of scale effects," Physical Chemistry Chemical Physics, 2024, 26(28), 19069–19082. DOI

3. K. E. Karim, B. Kim, "Molecular admission and packing govern transport in subnanometre graphene slits," Physical Chemistry Chemical Physics (2026), published online 9 October 2026. DOI

Manuscripts — in submission

4. K. E. Karim, B. Kim, "Wall–fluid interaction strength shifts the slit width of maximum finite-time diffusivity in a graphene-confined simple fluid," in submission.

5. K. E. Karim, B. Kim, "Confinement-Dependent Hydraulic Resistance and Limits of Navier-Slip Transferability in Graphene Nanoslits," in submission.

Manuscripts — in preparation

6. K. E. Karim, B. Kim, "Observable-Specific Identifiability of Energy, Occupancy Entropy, and Apparent Slip in Single-File CNT Transport," working title; in preparation.

7. K. E. Karim, B. Kim, "Physics-informed neural networks (PINNs) for a structure–friction–flow closure in nanoconfined water," in preparation. An analytic slip-flow solution carries the physics, the network learns only the residual that the theory misses, and models are scored only on geometries withheld from training.

Further directions: multimodal electrolyte transport in CNTs; error-aware MLIPs for adaptive simulation.

Additional peer-reviewed work

A. S. M. M. Rahman, K. E. Karim, M. H. S. Simanto, "Effect of heat treatment on low carbon steel: An experimental investigation," Applied Mechanics and Materials, 2016, 860, 7–12. DOI

Also: an RPR-manipulator journal article and five peer-reviewed conference papers (thermal, materials, design).

TECHNICAL EXPERTISE

- **Molecular simulation —** LAMMPS; EMD/NEMD; NVT/NPT/NVE; confined fluids; pressure-driven and Couette flow; Green–Kubo and NEMD transport coefficients; interfacial friction/slip; spatial binning and profile analysis.
- **Electronic structure & MLIP —** CP2K periodic GPW reference calculations using PBE-D3(BJ), GTH pseudopotentials and DZVP-MOLOPT basis sets; DP-GEN active learning; DeePMD-kit dataset curation, committee training, holdout validation, error analysis, stability testing and LAMMPS deployment.
- **Scientific computing & statistics —** Python (NumPy, SciPy, pandas, Matplotlib, MDAnalysis, scikit-learn), OVITO, VMD, Bash, Linux/HPC, MPI workflows, automated parameter sweeps; block averaging, confidence intervals, convergence/sensitivity diagnostics, uncertainty-aware regression.
- **Continuum modelling & CFD —** Analytic Hagen–Poiseuille and Navier-slip flow solutions; ANSYS meshing, boundary-condition set-up, compressible-flow and thermal analysis, field post-processing.

ADDITIONAL PROJECTS & EXPERIENCE

Thermo-Fluid Simulation of a Methane Rocket-Engine Nozzle & Cooling Passage 2022

Industry-linked laboratory project: ANSYS geometry/mesh generation, compressible nozzle-flow and high-temperature thermal analysis of the nozzle wall and methane coolant passage, with field post-processing.

Research Assistant — Robotics & Mechatronics Laboratory, KUET Feb 2014 – May 2015

Kinematic modelling of an RPR manipulator, automated-wheelchair design, and experimental thermal/materials studies.

TEACHING & MENTORING

Practical MLIP Crash Course (CP2K → DeePMD-kit → LAMMPS) — course developer 2026 – Present

- **Curriculum —** Developing a 30-lesson practical course for senior undergraduates and first-year M.S. students: runnable recipes, clearly labelled synthetic examples, a break-it-and-fix-it case in every lesson (worked solutions complete for six core lessons) and an interview playbook; learner pilot planned.

REFERENCES

References available upon request.